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Steering Language Models in Multi-Token Generation: A Case Study on Tense and Aspect

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Abstract

Large language models (LLMs) are able to generate grammatically well-formed text, but how do they encode their syntactic knowledge internally? While prior work has focused largely on binary grammatical contrasts, in this work, we study the representation and control of two multidimensional hierarchical grammar phenomena—verb tense and aspect—and for each, identify distinct, orthogonal directions in residual space using linear discriminant analysis. Next, we demonstrate causal control over both grammatical features through concept steering across three generation tasks. Then, we use these identified features in a case study to investigate factors influencing effective steering in multi-token generation. We find that steering strength, location, and duration are crucial parameters for reducing undesirable side effects such as topic shift and degeneration. Our findings suggest that models encode tense and aspect in structurally organized, human-like ways, but effective control of such features during generation is sensitive to multiple factors and requires manual tuning or automated optimization.[1]

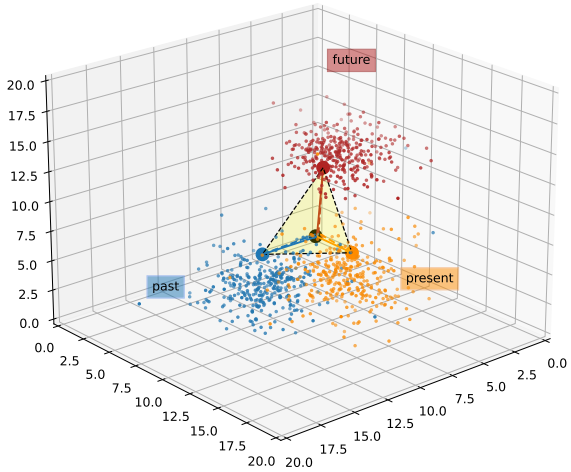

Figure 1: Activations from Qwen-7B projected along three identified feature directions that represent the categorical values of **tense**: present, past and future.

1 Introduction

Growing evidence on the generative capabilities of large language models (LLMs) suggests that they encode structural properties of language—such as syntax trees—within their hidden representations (Hewitt and Manning, 2019; Diego Simon et al., 2024). Studying the representation of these properties can reveal how syntactic structures in models compare to those in human language. It may also help develop more linguistically aware AI systems, extend model capabilities to low-resource languages, improve the interpretability of generation, and assist in diagnosing systematic errors in tasks like machine translation (López-Otal et al., 2025).

Prior work has assessed grammatical competence with behavioral evaluations, using black-box measures such as perplexity (Chen et al., 2024) and grammatical acceptability benchmarks (Warstadt et al., 2020; Hu et al., 2020). However, these methods do not reveal how grammatical concepts are internally represented. To address this, recent interpretability studies have examined the mechanisms underlying syntactic competence, such as indirect object identification (Wang et al., 2023) and subject-verb agreement (Ferrando and Costa-jussà, 2024), providing detailed insights into the role of specific architectural components. Yet, they largely focus on binary grammatical contrasts (e.g., singular vs. plural agreement), and often evaluate on single-token continuations, which limit their ability to capture more compositional or distributed grammatical phenomena.

In this work, we take a complementary approach focused on grammatical **tense** and **aspect**, two core verb properties that encode temporal relations and often span multiple tokens. While tense positions

[1]https://github.com/klerings/tense-aspect

an event in different time periods relative to the moment of speech (i.e., past, present or future),

a) She *drove* her car.
b) She *drives* her car.
c) She *will drive* her car.

aspect introduces an additional layer of temporal structure—describing whether an event is completed, ongoing or repeated (e.g., simple, perfect, progressive or perfect progressive) (Klein, 2009).

d) She *has driven* her car.
e) She *is driving* her car.
f) She *has been driving* her car.

Unlike binary syntactic features, tense and aspect involve multiple discrete categories and can be combined (e.g., past perfect). Theoretically, they are independent grammatical features: any aspect should, in principle, be expressible in any tense. This separability is useful for a mechanistic analysis, as it allows for a systematic exploration of how these categories interact. We locate and analyze the representation of both properties in two models via linear probes and Linear Discriminant Analysis (LDA) (Park et al., 2024a), and test their causal relevance (Vig et al., 2020; Mueller et al., 2024) by using them to steer LLM outputs in multi-token generation.

While prior work has demonstrated representational steering for single-token continuations and has focused on semantic properties, our approach applies steering to full sentences by intervening on every generated token individually, and focuses on grammatical phenomena. Since steering multiple tokens increases the risk for degenerate outputs and unintended side effects (Bricken et al., 2023; Stickland et al., 2024), we analyze when and how strongly to intervene. We find that steering effectiveness depends on activation norms, model architecture, and the nature of the task.

Contributions. Our key contributions are as follows:

- We identify distinct, near-orthogonal directions in LLM residual space encoding tense and aspect, using probes and LDA.
- We demonstrate causal control over these grammatical features through concept steering, thereby mediating the LLM output during multi-token generation.
- We investigate conditions for effective steering, analyzing the influence of steering strength and location, activation norm, model type and task on side effects.

2 Background and Related Work

Linear Representation Hypothesis and Concept Directions. The linear representation hypothesis is the assumption that abstract features are encoded as linear directions in LLM residual space (Elhage et al., 2022; Nanda et al., 2023; Park et al., 2024b; Marks and Tegmark, 2024; Park et al., 2024a). This has led to the notion of **feature directions**—unit vectors in activation space corresponding to specific properties—and, more generally, **feature subspaces**, which span sets of related directions (e.g., different tense directions in shared tense subspace) (Geiger et al., 2025; Mueller et al., 2024).

Such representations have been extracted using sparse autoencoders (Bricken et al., 2023; Huben et al., 2024), dimensionality reduction (Gurnee and Tegmark, 2024), (Heinzerling and Inui, 2024), and supervised linear probes (Marks and Tegmark, 2024). Recently, Park et al. (2024a) introduced an LDA-based method to identify scaled directions that capture categorical structure. Unlike prior approaches focused on concepts with natural opposites (e.g. MALE vs. FEMALE), this framework models categorical concepts (e.g., TENSE) as sets of **binary features** (e.g., FUTURE, PRESENT, PAST), independently of predefined class structures.

Steering Language Models. Feature steering has mainly targeted semantic features like sentiment (Rimsky et al., 2024; Lee et al., 2025) or numbers (Heinzerling and Inui, 2024), while categorical linguistic properties remain underexplored. Earlier work has focused on single-token interventions, with recent studies beginning to explore steering full-sentence generation (Lee et al., 2025; Rimsky et al., 2024; Wu et al., 2025). Systematic evaluations, e.g., of steering strength, side effects, or degeneration, are still sparse (Pres et al., 2024).

Our work addresses these gaps by steering categorical grammatical features—specifically, tense and aspect—during sentence-level generation with controlled intervention strength and position-aware application. We extend the LDA-based framework from Park et al. (2024a) by broadening its application from lexical to sentence-level grammatical categories.

Grammatical Knowledge in LMs. The grammatical competence of language models has been an ongoing research topic since before the emergence of LLMs, with subfields such as "Bertology" (Rogers et al., 2020) focusing on earlier architectures like BERT. Tense has been explored in behavioral evaluations as well as causal analyses (Merullo et al., 2024; Zhang et al., 2025), but typically via binary distinctions (e.g., past vs. present) and single-token interventions—aside from Brinkmann et al. (2025), who consider open-ended generation. However, tense and aspect have not been jointly analyzed in a unified framework. We address this by studying these features in multi-token generation, combining probing, representation space analysis, and causal steering to examine how grammatical concepts are encoded and can be controlled. For a more comprehensive review of related work, see App. B.

3 Locating Tense and Aspect

We focus on the interlinked grammatical concepts of verb tense and aspect by localizing and visualizing their subcategories via exploratory methods.

3.1 Experimental Setup

Tense and aspect are core grammatical categories that encode temporal structure. Tense refers to the time at which an event occurs (present, past, future), while aspect characterizes its temporal structure (simple, progressive, perfect, perfect progressive). In theory, each tense can occur in combination with each aspect, yielding a grid of 12 distinct tense-aspect forms (see App. C). This system makes these features ideal for studying multidimensional hierarchical structure in LLMs. In practice, such regular and compositional combinations are rare across languages. The comparatively structured tense-aspect system in English is an exception (Klein, 2009). Therefore, we focus our study on English rather than languages with less regularity.

Data. We analyze sentences annotated with grammatical tense and aspect, requiring each sentence to contain a single, unambiguous tense-aspect combination to isolate a clear signal for each target variable. We use sentences from the Penn Treebank (Marcus et al., 1993) annotated with PropBank (Kingsbury and Palmer, 2002), which provides verb-specific tense and aspect labels. After filtering out ambiguous sentences, the class distribution is highly skewed, with most examples in the simple aspect and some targets having fewer than ten instances. To address this imbalance, we augment rare classes with synthetic examples generated by GPT-4o (Hurst et al., 2024), resulting in 1,543 labeled sentences. Augmentation improves balance, but some categories remain overrepresented (e.g., simple past), therefore we downsample them for training the classifiers. For evaluation, we use the verb tense subset of 281 sentences from BIG-bench (Srivastava et al., 2023; Logeswaran et al., 2018) (see App. D for dataset details).

Models. We use the models Llama-3.1-8B-Instruct (Dubey et al., 2024), primarily trained on English data, and Qwen-2.5-7B-Instruct (Qwen et al., 2025), with a focus on English and Chinese, that are commonly studied for similar analyses. For brevity, we omit version numbers and the 'Instruct' suffix throughout the remainder of the paper.

Localization via Probing. To localize the target concepts—tense, aspect and their combination—we train linear probing classifiers (Belinkov, 2022) on the hidden representations of the pre-trained LLMs. Let the model be defined as $f_\theta : x \rightarrow h$, where θ are learnable parameters, $x = (x_1, ..., x_N)$ the input tokens, and $h^l = (h^l_1, ..., h^l_N)$ the hidden representations at layer l. At each layer, we train a probing model $p^l : h^l_{\text{agg}} \rightarrow y$ that maps aggregated hidden states h^l_{agg} to the corresponding tense, aspect or tense-aspect labels y using multinomial logistic regression. We compute h_{agg} per layer as follows,

$$h_{\text{agg}} = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} h_i, \tag{1}$$

summing token-level representations and normalizing by the square root of sequence length N, a strategy that outperforms other aggregation methods, see App. E for more details. Finally, we mean-center the aggregated hidden representations before classification.

Representation Geometry. To analyze how tense and aspect are represented in model activations, we use the framework proposed by Park et al. (2024a) for handling categorical features. In this approach, each categorical concept (e.g., TENSE) consists of a set of subordinate feature values (e.g., PAST, PRESENT, FUTURE). To model each categorical value as a direction, they are represented as **binary features**: {IS_PAST, IS_NOT_PAST}, {IS_PRESENT, IS_NOT_PRESENT}, {IS_FUTURE,